

KINETIC COMPARISONS DURING VARIATIONS OF THE POWER CLEAN

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ABSTRACT

Comfort, P, Allen, M, and Graham-Smith, P. Kinetic comparisons during variations of the power clean. *J Strength Cond Res* 25(12): 3269–3273, 2011—The aim of this investigation was to determine the differences in peak power, peak vertical ground reaction forces, and rate of force development (RFD) during variations of the power clean. Elite rugby league players ($n = 16$; age 22 ± 1.58 years; height 182.25 ± 2.81 cm; body mass 98.65 ± 7.52 kg) performed 1 set of 3 repetitions of the power clean, hang power clean, midhigh power clean, or midhigh clean pull, using 60% of 1 repetition maximum power clean, in a randomized order, while standing on a force platform. One-way analysis of variance with Bonferroni post hoc analysis revealed a significantly ($p < 0.001$) greater peak power output during the midhigh power clean ($3,565.7 \pm 410.6$ W) and the midhigh clean pull ($3,686.8 \pm 386.5$ W) compared with both the power clean ($2,591.2 \pm 645.5$ W) and the hang power clean ($3,183.6 \pm 309.1$ W), along with a significantly ($p < 0.001$) greater peak F_z during the midhigh power clean ($2,813.8 \pm 200.5$ N) and the midhigh clean pull ($2,901.3 \pm 226.1$ N) compared with both the power clean ($2,264.1 \pm 199.6$ N) and the hang power clean ($2,479.3 \pm 267.6$ N). The midhigh power clean ($15,049.8 \pm 4,415.7$ N·s⁻¹) and the midhigh clean pull ($15,623.6 \pm 3,114.4$ N·s⁻¹) also demonstrated significantly ($p < 0.001$) greater instantaneous RFD when compared with both the power clean ($8,657.9 \pm 2,746.6$ N·s⁻¹) and the hang power clean ($10,314.4 \pm 4,238.2$ N·s⁻¹). From the findings of this study, when training to maximize power, F_z , and RFD, the midhigh power clean and midhigh clean pull appear to be the most advantageous variations of the power clean to perform.

KEY WORDS peak force, instantaneous rate of force development, peak power

INTRODUCTION

Variations of the clean and power clean, incorporating different starting positions (from the floor, hang, and midhigh), are commonly incorporated into strength and conditioning programs. It has been suggested that such exercises increase an athlete's performance by imitating sport-specific movements, while concurrently using explosive power (15,16), with performance in the hang power clean being correlated to both 20-m sprint performance and countermovement jump performance (10).

The majority of research studies regarding the kinetic characteristics during performances of the power clean and its variations have focused on the load that achieves peak power output (3,4,11–13). Kawamori et al. (12) determined that peak power output, during the midhigh clean pull is achieved at 60% of 1-repetition maximum (1RM) (power clean), although the shortest time to peak rate of force development (RFD; 99.8 ± 14.0 milliseconds) was achieved at 30% of 1RM. Interestingly, time to peak RFD during midhigh clean pulls, at all loads (30, 60, 90, 120% of 1RM Power Clean), was shorter than time to peak RFD in both countermovement jumps (263.3 ± 63.5 milliseconds) and vertical jumps (194.7 ± 27.0 milliseconds). These findings indicate that midhigh clean pulls and possibly midhigh power cleans may be preferential, over vertical jump activities, when the focus of training is to improve RFD in an athlete. Previously, Kawamori et al. (11) found that peak power output during the hang power clean is achieved using a load of 70% of 1RM power clean. More recently, however, Kilduff et al. (13) found that peak power output during the hang power clean was not significantly different ($p > 0.05$) between loads of 50, 60, 70, 80, or 90% of 1RM power clean. During power cleans, however, Cormie et al. (3,4) demonstrated that peak power is achieved at a load of 80% 1RM.

Enoka (5) studied experienced weightlifters' technique during the pull phase of a clean and found that subjects created a peak ground reaction force (F_z) of 2,471 N during the first pull phase, whereas the second pull phase created a greater peak F_z with an average of 2,809 N. Häkkinen et al. (9) found similar results, with the second pull displaying the greatest peak F_z at 150% of the system load, with Garhammer (6–8) also identifying the second pull phase as eliciting the

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highest power output compared with the first pull across weight classes in Olympic weightlifters. More recently, Souzam et al. (14) also found that the second pull phase of the power clean resulted in a significantly higher peak F_z compared with the first pull, unweighting, and catch phases. To date, only one study has compared the peak F_z and RFD of the power clean (performed from the floor), the hang power clean (performed from the knee), the midhigh power clean (performed from the midhigh), and the midhigh clean pull (performed from the midhigh, without the catch). The authors found that peak F_z and RFD were significantly ($p < 0.001$) during the midhigh power clean ($2,801.7 \pm 195.4$ N; $14,655.8 \pm 4,535.1$ N·s⁻¹) and the midhigh clean pull ($2,880.2 \pm 236.2$ N; $15,320.6 \pm 3,533.3$ N·s⁻¹) compared with both the power clean ($2,306.2 \pm 240.5$ N; $8,839.7 \pm 2,940.4$ N·s⁻¹) and the hang power clean ($2,442.9 \pm 293.2$ N; $9,768.9 \pm 4,012.4$ N·s⁻¹) at a load of 60% 1RM power clean (2).

There is little evidence to indicate as to which technique variations may be optimal in terms of generating peak power, force, and RFD (5,9,14). The majority of available evidence has determined that the second pull phase of the power clean results in the greatest vertical ground reaction forces (F_z) (5,9,16), with one study identifying that the midhigh power clean and midhigh pull result in a significantly higher peak F_z and RFD compared with other variations of the power clean (2). Therefore, the aim of this study was to compare peak power, peak F_z , and RFD during the power clean, hang power clean, midhigh power clean, and midhigh pull. It was hypothesized that the midhigh variations would result in a higher peak power output because they have previously been shown to elicit higher peak F_z and instantaneous RFD when compared with the power clean and hang power clean.

METHODS

Experimental Approach to the Problem

This study employed a within-subjects repeated measures research design, whereby peak power output was determined during the power clean, hang power clean, midhigh power clean, and midhigh clean pull. Peak power, peak F_z , and RFD were measured by the athlete performing all exercise variations while standing on a force platform (Kistler, Winterthur, Switzerland, Model 9286AA, SN 1209740).

Subjects

Sixteen healthy male rugby league players ($n = 16$; age 22 ± 1.58 years; height 182.25 ± 2.81 cm; body mass 98.65 ± 7.52 kg) participated in this study. All participants had regularly performed structured strength and conditioning training, including variations of the clean, for ≥ 2 years. The investigation was approved by the Institutional Ethics Committee, and all the subjects provided informed consent before participation. The study conformed to the principles of the World Medical Association's Declaration of Helsinki. The participants had previously conducted technique sessions,

supervised by a certified strength and conditioning coach, within their normal training to allow familiarization with the protocols and ensure appropriate technique. Testing took place during preseason training, while the participants were midway in the taper phase of a power mesocycle.

Testing

Before testing, all subjects performed a standardized dynamic warm-up, including each variation of the power clean (4 repetitions, 3 sets) using a standardized load (40 kg) (Ivanco Olympic bar and weights, Hialeah, Florida, USA). The participants were then randomly assigned to perform 1 cluster set of 3 repetitions (30 seconds of rest between repetitions to minimize fatigue) of each exercise, starting with either the power clean (bar starting midway up the shin and caught in a shallow squat), hang power clean (bar starting in line with the top of the patella and caught in a shallow squat), midhigh power clean (bar starting in line with the middle of the thigh and caught in a shallow squat), or midhigh clean pull (bar starting in line with the middle of the thigh, but no catch phase). All lifts were performed using a standardized load of 60% of each individual's previously determined 1RM power clean to minimize technical errors associated with higher loads and to ensure that the results were comparable with the findings of previous research (2). One repetition maximum power cleans were assessed on 2 separate occasions, 3–5 days apart, to determine reliability after a standardized protocol (1).

All lifts were performed with the subjects standing on a force plate (Kistler, Model 9286AA, SN 1209740), sampling at 1,000 Hz, interfaced with a laptop. Data were later analyzed using Bioware (Version 3.22; Kistler Instrument Corporation) to determine peak vertical ground reaction force. Instantaneous RFD was determined by dividing the difference in consecutive vertical force readings by the time interval (0.001 seconds) between readings. Data were smoothed using a moving average window of 400 milliseconds. Instantaneous peak power was determined from the product of the vertical ground reaction force and vertical velocity. The velocity of the center of mass was determined by the integration of the acceleration data (derived from Newton's second law), whereby the starting velocity was zero. To calculate power in this way it was important that the initial vertical ground reaction force represented system load (athlete's body mass plus load lifted). Consequently, the bar was held slightly above the ground level before the onset of the full power clean.

Statistical Analyses

A 1-way analysis of variance and Bonferroni post hoc analysis were conducted to determine if there were any significant differences in peak power output between lifts. Intraclass correlation coefficients were calculated to determine the reliability between 1RM power cleans and to establish reproducibility between the repetitions during each exercise variation. Statistical power was calculated between 0.89 and 0.92 for each lift. A priori alpha level was set to $p \leq 0.05$.

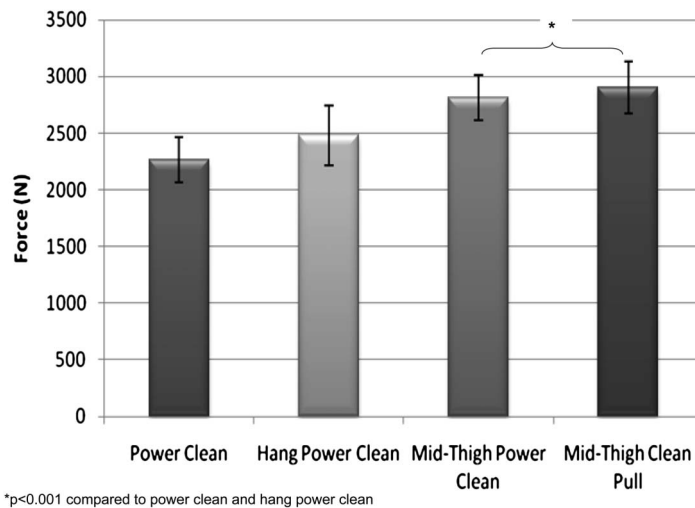


Figure 1. Comparison of peak F_z between variations of the power clean.

RESULTS

Intraclass correlation coefficients demonstrated a high and statistically significant level of reliability for the 1RM power clean ($r=0.96$, $p<0.001$) and a high level of reproducibility between repetitions for the power clean ($r=0.92$, $p<0.01$), hang power clean ($r=0.98$, $p<0.001$), midthigh power clean, and midthigh clean pull ($r=0.98$, $p<0.001$).

One-way analysis of variance demonstrated significant differences ($p<0.001$) in peak vertical ground reaction forces between variations of the power clean. Bonferroni post hoc

analysis revealed a significantly ($p<0.001$) greater peak F_z during the midthigh power clean (2801.7 ± 195.4 N) and the midthigh clean pull (2880.2 ± 236.2 N) compared with both the power clean (2306.2 ± 240.5 N) and the hang power clean (2442.9 ± 293.2 N) (Figure 1).

No significant ($p>0.05$) differences were found when comparing the peak F_z between the midthigh power clean (2813.82 ± 200.5 N) and the midthigh clean pull (2901.3 ± 226.1 N). There were no significant differences in peak F_z between the hang power clean (2479.3 ± 267.8 N) and the power clean (2264.1 ± 199.6 N) (Figure 1).

Significant differences ($p<0.001$) in instantaneous RFD were found between variations of the power clean. Bonferroni post hoc analysis revealed a significantly ($p<0.001$) greater RFD during the midthigh power clean ($15,049.8 \pm 4,415.7$ N·s⁻¹) and the midthigh clean pull ($15,623.6 \pm 3,114.4$ N·s⁻¹) compared with both the power clean ($8,675.8 \pm 2,746.6$ N·s⁻¹) and the hang power clean ($10,314.4 \pm 4,238.2$ N·s⁻¹) (Figure 2).

No significant ($p>0.05$) differences were found when comparing the RFD between either the midthigh power clean ($14,655.8 \pm 4,535.1$ N·s⁻¹) and the midthigh clean pull ($15,320.6 \pm 3,533.3$ N·s⁻¹), or between the hang power clean ($9,768.9 \pm 4,012.4$ N·s⁻¹) and the power clean ($8,839.7 \pm 2,940.4$ N·s⁻¹) (Figure 2).

Significant differences ($p<0.001$) in peak power output were found between variations of the power clean. Bonferroni post hoc analysis revealed a significantly ($p<0.001$) greater peak power output during the midthigh power clean ($3,565.7 \pm 410.6$ W) and the midthigh clean pull ($3,686.8 \pm 386.5$ W) compared with both the power clean ($2,591.2 \pm 645.5$ W) and the hang power clean ($3,183.6 \pm 309.1$ W). However, no significant ($p>0.05$) differences were found when comparing the peak

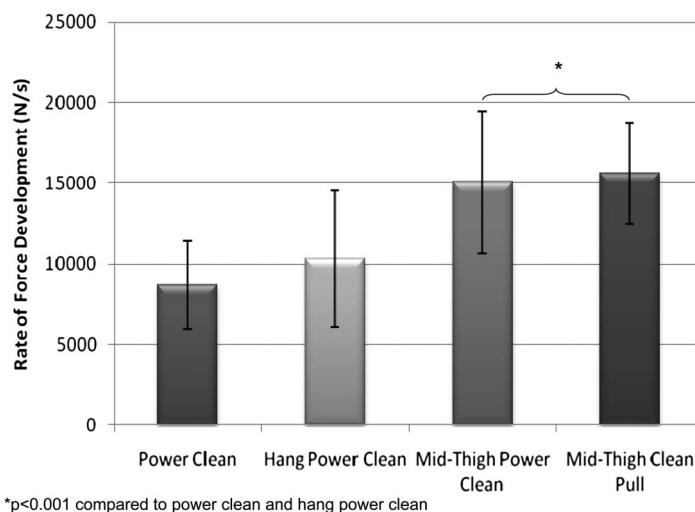


Figure 2. Comparison of peak rate of force development (RFD) between variations of the power clean.

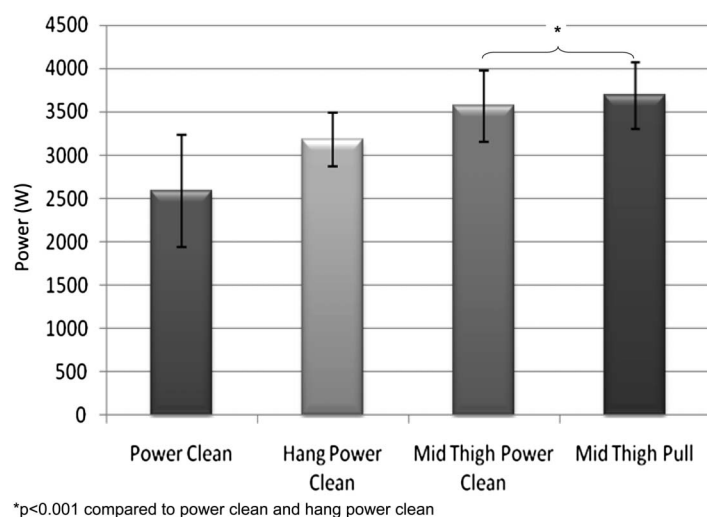


Figure 3. Comparison of peak power output during variations of the power clean.

power output between either the midhigh power clean ($3,565.7 \pm 410.6$ W) and the midhigh clean pull ($3,686.8 \pm 386.5$ W), and between the hang power clean ($3,183.6 \pm 309.1$ W) and the power clean ($2,591.2 \pm 645.5$ W) (Figure 3).

DISCUSSION

The findings of this study should help strength and conditioning coaches make evidence-based decisions of the most beneficial variations of the clean to perform during different phases of a periodized training program.

The results of this study demonstrate significant differences ($p < 0.001$) in peak power output between variations of the power clean, with significantly ($p < 0.001$) greater peak power output during the midhigh power clean ($3,565.7 \pm 410.6$ W) and the midhigh clean pull ($3,686.8 \pm 386.5$ W) compared with both the power clean ($2,591.2 \pm 645.5$ W) and the hang power clean ($3,183.6 \pm 309.1$ W). No significant differences ($p > 0.05$) were found when comparing the peak power output between either the midhigh power clean ($3,565.7 \pm 410.6$ W) and the midhigh clean pull ($3,686.8 \pm 386.5$ W), or between the hang power clean ($3,183.6 \pm 309.1$ W) and the power clean ($2,591.2 \pm 645.5$ W). These trends are in line with prior research findings that demonstrated higher F_z and RFD during the midhigh power clean and midhigh clean compared with the power clean and hang power clean (2). Such findings also support previous findings, which demonstrate that the second pull phase of the clean generates the greatest power output (6–8).

The results also showed greater peak F_z during the midhigh power clean ($2,801.7 \pm 195.4$ N) and the midhigh clean pull ($2,880.2 \pm 236.2$ N) compared with both the power clean ($2,306.2 \pm 240.5$ N) and the hang power clean

($2,442.9 \pm 293.2$ N). These values are in line with previous research that found significantly ($p < 0.001$) greater peak F_z during the midhigh power clean ($2,801.7 \pm 195.4$ N) and the midhigh clean pull ($2,880.2 \pm 236.2$ N) compared with both the power clean ($2,306.2 \pm 240.5$ N) and the hang power clean ($2,442.9 \pm 293.2$ N) (2). This also supports findings from previous research, which has demonstrated that the second pull phase of the clean results in greater F_z compared with the other phases of the lift (3,4,13). The F_z during the power clean ($2,306.2 \pm 240.5$ N) and midhigh power clean ($2,801.7 \pm 195.4$ N) were also comparable

with F_z during the first pull (2,471 N) and second pull (2,809 N) of the clean, respectively (5).

Comparisons of instantaneous RFD demonstrate similar trends to the power and force findings. The RFD was significantly ($p < 0.001$) greater during the midhigh power clean ($15,049.8 \pm 4,415.7$ N·s⁻¹) and the midhigh clean pull ($15,623.6 \pm 3,114.4$ N·s⁻¹) compared with both the power clean ($8,675.8 \pm 2,746.6$ N·s⁻¹) and the hang power clean ($10,314.4 \pm 4,238.2$ N·s⁻¹). However, there were no significant differences ($p > 0.05$) in RFD between the midhigh variations or between the power clean and hang power clean. These values and trends are in line with those of prior research that demonstrated significantly ($p < 0.001$) greater instantaneous RFD ($14,655.8 \pm 4,535.1$ and $15,320.6 \pm 3,533.3$ N·s⁻¹, respectively) than both the power clean ($8,839.7 \pm 2,940.4$ N·s⁻¹) and the hang power clean ($9,768.9 \pm 4,012.4$ N·s⁻¹), at a load of 60% 1RM power clean (2).

The similarity in peak power, peak F_z and the instantaneous RFD between the midhigh power clean and the midhigh clean pull may be explained by the fact that the concentric phases of the lifts are kinematically identical, whereas the hang power clean and power clean exhibit noticeably different kinematics as previously hypothesized by Comfort et al (2). These higher values during the midhigh variations may also be obtained as a result of the reduced bar displacement, compared with the other variations, resulting in a greater requirement for acceleration of the bar, especially in the case of the midhigh power clean, where the bar needs to be caught on the anterior deltoids. The midhigh power clean and the midhigh clean pull demonstrate higher peak power output, F_z and higher instantaneous RFD when compared with the hang power clean and the power clean. The midhigh variations of the clean may also provide further advantages in terms of improving RFD

as Kawamori et al. (12) found that time to peak RFD during midhigh clean pulls, at all loads (30, 60, 90, 120% of 1RM Power Clean) was shorter (Peak at 30% 1RM, 99.8 ± 14.0 milliseconds) than time to peak RFD in both vertical jumps (194.7 ± 27.0 milliseconds) and countermovement jumps (263.3 ± 63.5 milliseconds). These findings and the findings of the present study indicate that midhigh clean variations may be preferential, over vertical jump activities, when focusing on improving the RFD in an athlete.

Hori et al. (10) found that performance in the hang power clean is related to both 20-m sprint performance and countermovement jump performance; however, no research has made such comparisons using the midhigh power clean and midhigh clean pull. These exercises may demonstrate a stronger relationship to such performances based on the F_z and instantaneous RFD findings of this study. Furthermore, research has demonstrated that faster top running speed are achieved through greater peak F_z rather than the development of limb speed (17–19); therefore, the use of midhigh power clean variations to develop F_z may assist with the development of sprint speed. Research into the relationship between peak F_z during midhigh power clean variations and top running are recommended.

It is recommended that further research determine if these findings are similar at different loads, or if F_z and instantaneous RFD are greater during the hang power clean and power clean when performed at the loads that have previously been reported to result in peak power output (70 and 80% 1RM, respectively) (3,4,10,12).

PRACTICAL APPLICATIONS

From the results of this study, the most advantageous variations of the clean, when training to maximize power output, F_z , and RFD, appear to be the midhigh power clean and midhigh pull. These variations also offer the practical benefit that both are easy for less experienced athletes to learn and require less technical excellence.

During a strength-based Mesocycle, it is suggested that the midhigh clean pull be used because this can be performed at loads $>100\%$ 1RM power clean, because the catch phase is not required. During a power-based Mesocycle, it is suggested that the midhigh power clean may be advantageous because it results in higher peak power output, peak F_z , and RFD compared with the other variations of the clean.

Further research investigating performance in these lifts, ideally incorporated into a training study, and their relationships to sprint and jump performances are also recommended.

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